

Indicators and Predictors of the Suspension of Disbelief: Children's Individual Presence Tendencies

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ABSTRACT

Presence is a phenomenon where a person distributes his/her attention to internal (mental) or distal (sensory) cues. Research shows, that not only technological immersion, but also person-specific factors, influence the sense of presence. These factors may cause an individual presence tendency (IPT) that affects how presence is experienced. This paper investigates if an IPT exists, how it can be calculated, and what factors influence it. A study is presented with 78 participants, aged 13-16, who experienced three different environments within different immersive settings. The results show that level of technological immersion has a strong effect on presence.

Index Terms: Human-centered computing—HCI theory, concepts and models; Human-centered computing—Virtual reality; Human-centered computing—User studies

1 INTRODUCTION

Presence has always been seen as a key characteristic of creating virtual realities. A lot of research focuses on the technological characteristics of Virtual Reality (VR) experiences and their effects on presence. As such effects are already well documented, this paper investigates the person-specific key characteristic of immersive experiences: The suspension of disbelief. It is known that the same level of immersion tends to affect people differently, depending on their willingness to accept the VR as an actual reality. In order to assess the user's individual presence tendency (IPT), a study with 78 secondary school students is presented. This study compares three comparable virtual environments, that were each presented using three different devices. A comparison of the users' presence reports between the different technologies and provided environments investigates if there is an IPT and how it can be measured.

2 FACTORS INFLUENCING PRESENCE

Many researchers use the definition of presence as the feeling of 'being there' ([5]). As a sense of 'being there' suggests a place and refers mostly to physical presence, this paper prefers the definition of Lombard and colleagues of presence being the "perceptual illusion of nonmediation" [8, p. 77], regarding physical space, social actors, and representations of the user's self (for detailed discussions about the different types of presence, see e.g. Biocca [1]).

Immersion describes "a quantifiable description of technology" [12]. Based on studies comparing different technological devices (regarding their field of view [7] or how the avatar is represented [9]) regarding their effect on presence, it can be concluded that immersion has an impact on presence.

While the provided level of immersion can be seen as the strongest predictor of the sense of presence, subjective factors influence presence as well [3]. The suspension of disbelief, as a person-specific

factor, describes the users willingness to accept the VR as his/her actual reality. Slater and Usoh give a description of the willing suspension of disbelief where "the person truly 'knows' that s/he is in one place while experiencing inputs to his/her sense organs that to some extent provide the experience of being somewhere else" [10]. This leads to the feeling that "you are there" in the virtual reality [8], which is central to the experience of presence. There is some discussion about terminology in terms of VR [6] and suspension of disbelief is sometimes confused with presence itself, but it is actually a different construct [13], which influences presence. While being an important subjective factor contributing to presence [2], the suspension of disbelief cannot be measured directly and assessing it objectively seems to be even more problematic than assessing the user's level of presence. In more specific terms, the suspension of disbelief in terms of VR can be described as the tendency of a person to experience presence at a given level of immersion. This can also be seen as an individual presence tendency (IPT). In a broader understanding, the IPT gathers all person-specific factors (such as emotion, motivation, and cognition, see [4]) including stable personality traits and unstable temporary states of the user.

Following these theoretical assumptions, the hypotheses H1) *The level of technological immersion predicts the level of presence* and H2) *A user's levels of presence correlate between (a) the different provided technologies and (b) environments* are investigated.

3 METHOD

3.1 Sample

A study with 78 secondary school students (36 f) from an Austrian school was conducted. The students were aged between 13 and 16.

3.2 Instruments

Three different technologies were selected as different levels of immersion: A laptop setting (16-inch monitor, Intel HD Graphics 5500) was considered to be the least immersive setting, followed by a Mobile VR setting (a Moto Z smartphone in a Google Daydream headset with a Bluetooth controller). An HTC Vive setting was thought of being the most immersive setting.

Three virtual environments for learning Computer Science were provided: *Bill's Computer Workshop* lets the user shrink and explore the inside of a computer by finding/repairing hardware components (CPU, RAM, hard drive). *Fluxi's Cryptic Potions* explains the key principle of asymmetric cryptography by using public and private potions for encrypting and decrypting letters in a medieval setting. *Pengu's Treasure Hunt* is a metaphorical representation of finite state machines where states are displayed as islands and transitions functions as boats between the islands. In all environments, the user used a point-and-click system in order to teleport and to interact with the environment (e.g. for picking up components or for talking to Bill). The interaction was the same for all devices. All three settings used the center of the field of view (FOV) as a cursor for selection.

The Slater-Usoh-Steed questionnaire [11] for measuring physical presence on a seven-point Likert scale was translated to German and adapted for three environments *Bill's Computer Workshop* ($m = 4.41$, $sd = 1.49$, $\alpha = .87$) *Fluxi's Cryptic Potions* ($m = 4.14$, $sd = 1.56$, $\alpha = .91$), and *Pengu's Treasure Hunt* ($m = 3.96$, $sd = 1.99$, $\alpha = .92$).

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	Laptop	Mobile VR	HTC Vive
Laptop	1.00		
Mobile VR	0.34 ^b	1.00	
HTC Vive	0.25 ^a	0.47 ^b	1.00

Note: ^a $p < .05$ ^b $p < .01$

Table 1: Correlations of Presence Scores in the Different Settings.

3.3 Procedure

Right before the study, the students were assigned to one of three groups determining which environment they would experience using which technology. Then, the participants were provided with the first technology (randomized within the group) and the first environment. After the VR experience, the student filled out the corresponding presence questionnaire. The students then continued with the next technology/environment, etc. During the study, there were several dropouts due to motion sickness and classtime restrictions. The final sample sizes were 71, 68 and 74 (213 data sets in total).

3.4 Findings

The differences between the presence means of the different environments were not significant for any of the provided technologies ($p < .05$), indicating that the environments themselves can be treated equally regarding their ability to induce a sense of presence. When comparing the overall standardized presence values between the technologies (see Tab. 1), an ANOVA shows significant differences ($F(2, 212) = 35.31, p < .0005$). Gabriel post-hoc tests (used because of slightly different sample sizes due to dropouts) verified significant differences between all settings ($p < .0005$ for laptop/Mobile VR and laptop/HTC Vive; $p < .01$ for Mobile VR/HTC Vive).

The individual presence levels show a moderate correlation between the laptop setting and the Mobile VR setting ($r = .34, p < .01$) and between the Mobile VR and the HTC Vive setting ($r = .47, p < .01$). There was a low correlation between the laptop setting and the HTC Vive setting ($r = .25, p < .05$), see (see 1. The individual presence levels also correlate between the different environments: Presence scores in Bill's Computer Workshop show a small correlation with Fluxi's Cryptic Potions ($r = .23, p = .06$, approaching significance) and Pengu's Treasure Hunt ($r = .14, p > .05$, n.s.). A small correlation between Fluxi's Cryptic Potions and Pengu's Treasure Hunt approached significance as well ($r = .24, p < .05$).

3.5 Discussion and Limitations

The results show that technological immersion is a strong predictor of presence (H1), aligning with many research efforts concerning the effect of the used medium on user experience done in the past.

The participant's levels of presence correlated significantly between the different technologies (supporting H2a). Correlations between the perceived level of presence in the different environments approached significance (H2b could not be supported). Still, these results indicate that each child had their own IPT. It can be assumed that the correlations were low to moderate because the users played different environments on the different technologies and vice versa.

There are some limitations to this study: Presence, a highly subjective construct, was measured using post-experience questionnaires. While easy to use and non-obtrusive, subjective rating scales are always prone to bias and misjudgment of participants. It also has to be noted that the reported study focused on physical presence and on children in the age of secondary school students. Any generalization of the results might be biased due to the age group. The sample size of 78 students, each providing three different presence scores, can be considered sufficient for the first indications of the correlation analysis, but was small for the regression model. For future studies, it can be interesting to have a fourth environment that is not used for calculating the IPT mean score but rather for predictions of presence in this environment using this score.

4 CONCLUSION

This article investigated the so-called suspension of disbelief as an individual presence tendency (IPT). Based on theoretical assumptions it can be said that the IPT is a subconscious characteristic influenced by a set of person-specific traits, states, and cognitive correlates, like motivation, the users emotional state and their cognitive abilities. A study was conducted to check whether there is a measurable and comparable IPT between individuals using their presence scores in different environments using different technologies and to analyze potential measurement methods. The study suggests that there is an IPT that spans across technologies and environments. Even though this study cannot provide a solid measurement instrument or a clear set of predictive factors, it showed first correlates with this highly-subjective factor. This work was able to show indicators on how it might be possible to grasp a complicated subjective construct such as the suspension of disbelief as an individual presence tendency, by combining the individually reported levels of presence in different environments using different technologies. Also, motivational traits, emotional states, and cognitive abilities can be seen as predictors of users' individual presence tendencies. The measurement method of including several presence measures to calculate an overall IPT score as a latent construct seems to have merit and improvements in presence measurement methods could contribute to creating a solid instrument for assessing participants' individual IPT scores.

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